

Solutions Analysis II

1. Suppose the function f is defined for all $x \in [-1.5, 2.5]$ by $f(x) = x^5 - 5x^3$.

(a) Determine for which values of x the value of the function is equal to zero.

$$\begin{aligned}x^5 - 5x^3 &= 0 \\x^5 &= 5x^3 \\x^2 &= 5 \\x &= \pm\sqrt{5}\end{aligned}$$

From the second equation we see that $x = 0$ is a possible solution. For $x = \pm\sqrt{5}$ we have to check whether these points are in our domain. This is true for $x = \sqrt{5}$, but not for $x = -\sqrt{5}$. Thus, the function has two roots.

(b) Calculate $f'(x)$ and find the extreme points of f . What is the maximum/the minimum of the function.

$f'(x) = 5x^4 - 15x^2$. The FOC gives us.

$$\begin{aligned}5x^4 - 15x^2 &= 0 \\5x^4 &= 15x^2 \\x^2 &= 3 \\x &= \pm\sqrt{3}\end{aligned}$$

When checking for the domain, we find that $x = 0$ and $x = \sqrt{3}$ serve as possible extreme points. Now we need to check the SOC.

$$\begin{aligned}f''(x) &= 20x^3 - 30x \\f''(x=0) &= 0 \\f''(x=\sqrt{3}) &= 30\sqrt{3} > 0\end{aligned}$$

At $x = 0$ we have a saddle point. At $x = \sqrt{3}$ there is a minimum.

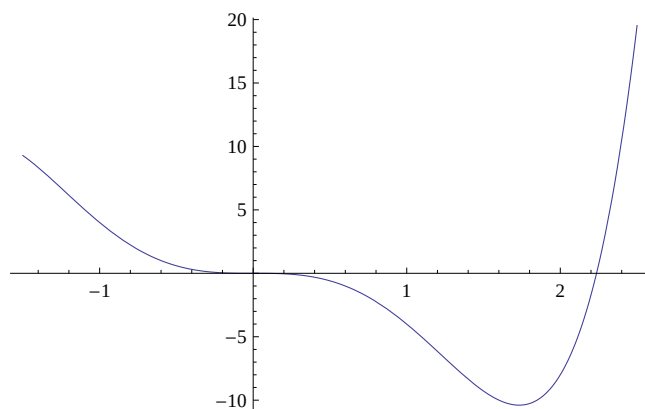
Are there any other minima/maxima? Yes, of course. We have to consider the boundaries of our domain. Both at $x = -1.5$ and $x = 2.5$ we have additional maxima.

The overall maximum of the function is attained at $x = 2.5$ with $f(x) \approx 19.5$. The overall minimum is $x = \sqrt{3}$ with $f(x) \approx -10.4$.

- (c) Does the function have inflection points?

Yes, it does. We already found the first inflection point, which also happens to be a saddle point.

We find the additional inflection points by setting $f''(x) = 0$. This gives $x = \pm\sqrt{1.5}$.



2. Which of the following functions of x are convex? Which are concave?

(a) $f(x) = (2x - 1)^6$
 $f'(x) = 6(2x - 1)^5 \cdot 2$
 $f''(x) = 5 \cdot 12(2x - 1)^4 \cdot 2 \geq 0 \implies \text{convex}$

- (b) $f(x) = 5x + 7$
The function is both convex and concave since the sets of points above and below the function are convex.

(c) $f(x) = x^5$
 $f'(x) = 5x^4$
 $f''(x) = 20x^3$
The function as a whole is neither convex nor concave (but we can specify this for parts of the function).

(d) $f(x) = \sqrt{1 + x^2}$
 $f'(x) = x(1 + x^2)^{-\frac{1}{2}}$
 $f''(x) = (1 + x^2)^{-\frac{1}{2}} + x^2(1 + x^2)^{-\frac{3}{2}} > 0 \implies \text{strictly convex}$

(e) $f(x) = x^5$ for $x \geq 0$
 $f''(x) = 20x^3 \geq 0 \forall x \geq 0 \implies \text{convex}$

(f) $f(x) = 5x^2 - x^4$ for $x \geq 1$
 $f'(x) = 10x - 4x^3$
 $f''(x) = 10 - 12x^2 < 0 \forall x \geq 1 \implies \text{strictly concave}$

3. Appeasement Problem (Ashworth and Bueno de Mesquita, 2006). For full text see exercise set.

- (a) Take the derivative with respect to x , set up the FOC, and solve for x .

$$\begin{aligned} 1 - 2x - q &= 0 \\ \frac{x^*(q)}{2} &= \frac{1 - q}{2} \end{aligned}$$

$x^*(q)$ represents state S's optimal choice of appeasement as a function of S's perceived military strength.

- (b) We can find comparative statics by examining how this equilibrium offer ($x^*(q)$) changes when q changes. Differentiating $x^*(q)$ with respect to q yields:

$$\frac{\partial x^*(q)}{\partial q} = -\frac{1}{2} < 0$$

Not surprisingly, the optimal offer is decreasing in q . The stronger S is militarily, the less willing S is to appease D.

4. A government has to decide about the allocation of its budget. Let x denote the share of the budget used for military and y the share of the budget used for social expenditures. The government has to use of all its budget and has the following utility function:

$$u(x, y) = e^{2x} + e^{2y}$$

Solve the government's optimization problem.

The governments optimization problem is:

$$\begin{aligned} \max_{x,y} e^{2x} + e^{2y} \text{ s.t.} \\ x + y = 1 \end{aligned}$$

Setting up the Langrangian yields:

$$\mathcal{L} = e^{2x} + e^{2y} - \lambda(x + y - 1)$$

Taking the partial derivatives with respect to x , y together with the budget constraint gives:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x} &= 2e^{2x} - \lambda \\ \frac{\partial \mathcal{L}}{\partial y} &= 2e^{2y} - \lambda \\ x + y &= 1 \end{aligned}$$

Setting the first and the second equation equal yields:

$$\begin{aligned} 2e^{2x} - \lambda &= 2e^{2y} - \lambda \\ x &= y \end{aligned}$$

Together with the budget constraint we know that, $x = y = \frac{1}{2}$.

5. Consider the function $f(x) = (x^2 + 2x)e^{-x}$.

- (a) Determine for which values of x the value of the function is equal to zero. We have to set $(x^2 + 2x)e^{-x} = 0$. We know that $e^{-x} > 0 \forall x \in \mathbb{R}$. Thus,

$$\begin{aligned} x^2 + 2x &= 0 \\ x &= \frac{-2 \pm \sqrt{4 - 0}}{2} = -1 \pm 1 \end{aligned}$$

The roots of the function are $x = -2$ and $x = 0$.

- (b) Calculate $f'(x)$ and find the extreme points of f . What is the maximum/the minimum of the function?

$$\begin{aligned}f(x) &= (x^2 + 2x)e^{-x} \\f'(x) &= -(x^2 + 2x)e^{-x} + (2x + 2)e^{-x} \\f''(x) &= (x^2 + 2x)e^{-x} - 2(2x + 2)e^{-x} + 2e^{-x}\end{aligned}$$

We take the FOC $f'(x) = 0$ to look for stationary points.

$$\begin{aligned}-(x^2 + 2x)e^{-x} + (2x + 2)e^{-x} &= 0 \\-(x^2 + 2x) + (2x + 2) &= 0 \\-x^2 + 2 &= 0 \\x &= \pm\sqrt{2}\end{aligned}$$

We have stationary points at $x = \pm\sqrt{2}$. We now have to check the SOC.

$$\begin{aligned}f''(x) &= (x^2 + 2x)e^{-x} - 2(2x + 2)e^{-x} + 2e^{-x} \\&= (x^2 + 2x - 2x - 2 - 2x - 2 + 2)e^{-x} \\&= (x^2 - 2x - 2)e^{-x}\end{aligned}$$

Again, we know that $e^{-x} > 0 \forall x \in \mathbb{R}$, so that we only have to consider the polynomial. For $x = -\sqrt{2}$, the polynomial $(2 + 2\sqrt{2} - 2) > 0 \implies$ local minimum.

For $x = \sqrt{2}$, the polynomial $(2 - 2\sqrt{2} - 2) < 0 \implies$ local maximum.

As the domain is not limited, we have to check for the limit of $f(x)$ for $x \rightarrow \pm\infty$ in order to specify whether the local extreme points are also global.

$$\begin{aligned}\lim_{x \rightarrow -\infty} (x^2 + 2x)e^{-x} &\approx e^{-x} = \infty \\ \lim_{x \rightarrow \infty} (x^2 + 2x)e^{-x} &\approx e^{-x} = 0\end{aligned}$$

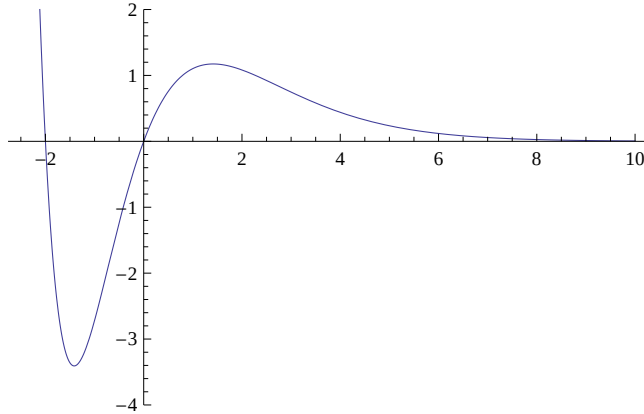
Therefore, the function does not have a global maximum. However, it has a global minimum since $f(-\sqrt{2}) < \lim_{x \rightarrow \infty} f(x)$.

- (c) Does the function have inflection points?

Yes, the function does have inflection points.

$$\begin{aligned}f''(x) &= 0 \\(x^2 + 2x)e^{-x} - 2(2x + 2)e^{-x} + 2e^{-x} &= 0 \\(x^2 + 2x) - 2(2x + 2) + 2 &= 0 \\x^2 - 2x - 2 &= 0 \\x &= \frac{2 \pm \sqrt{4 + 8}}{2} \\x &= 1 \pm \sqrt{3}\end{aligned}$$

- (d) Sketch the function and specify whether it is convex/concave (in sections).



The function is neither convex nor concave as a whole.

For concavity/convexity in parts of the function the inflection points are crucial.

We can see from the graph that the function is convex for all

$x \in (-\infty, 1 - \sqrt{3}]$ and $x \in [1 + \sqrt{3}, \infty)$.

It is concave for all $x \in [1 - \sqrt{3}, 1 + \sqrt{3}]$.

6. Derivate the indefinite integrals:

(a) $\int \frac{1}{\sqrt{x}} dx = 2\sqrt{x} + C$

(b) $\int e^{-4t} dt = -\frac{1}{4e^{4t}} + C$

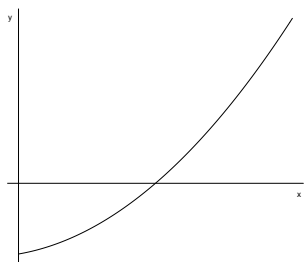
(c) $\int x\sqrt{x} dx = \frac{2}{5}x^{\frac{5}{2}} + C$

(d) $\int \frac{1}{x} = \ln x dx + C$

(e) $\int (2x^2 + x - 3) dx = \frac{2}{3}x^3 + \frac{1}{2}x^2 - 3x + C$

(f) $\int \frac{(x^4 + 1)^2}{x^3} dx = \frac{1}{6}x^6 + x^2 - \frac{1}{2x^2} + C$

7. Calculate $\int_0^2 (2x^2 + x - 3) dx$. Hint: Make a sketch of the function before.



At $x = 1$, the curve crosses the horizontal axis. In order to compute the total area "under the curve", compute $\int_0^1 (2x^2 + x - 3) dx + \int_1^2 (2x^2 + x - 3) dx$. Using the result from 1e, we can write:

$$\begin{aligned} \int_0^1 (2x^2 + x - 3) dx + \int_1^2 (2x^2 + x - 3) dx &= \left| \frac{2}{3}x^3 + \frac{1}{2}x^2 - 3x - (0) \right|_0^1 + \\ &\quad \left| \frac{2}{3}x^3 + \frac{1}{2}x^2 - 3x - \left(\frac{2}{3} + \frac{1}{2} - 3 \right) \right|_1^2 \\ &= \frac{11}{6} + \frac{4}{3} + \frac{11}{6} \\ &= 5 \end{aligned}$$

1.

$$\int \frac{1}{\sqrt{x}} dx = \int x^{-1/2} dx = \frac{x^{(-1/2)+1}}{(-1/2)+1} + C = \frac{x^{1/2}}{1/2} + C = 2\sqrt{x} + C.$$

2.

$$\int e^{-4t} dt \stackrel{u=-4t}{=} \int e^u \frac{du}{-4} = -\frac{1}{4} \int e^u du = -\frac{1}{4}e^u + C = -\frac{1}{4}e^{-4t} + C.$$

3.

$$\int x\sqrt{x} dx = \int x \cdot x^{1/2} dx = \int x^{3/2} dx = \frac{x^{(3/2)+1}}{(3/2)+1} + C = \frac{x^{5/2}}{5/2} + C = \frac{2}{5}x^{5/2} + C.$$

4.

$$\int \frac{1}{x} dx = \ln |x| + C.$$

5.

$$\begin{aligned} \int (2x^2 + x - 3) dx &= 2 \int x^2 dx + \int x dx - 3 \int 1 dx \\ &= 2 \cdot \frac{x^3}{3} + \frac{x^2}{2} - 3x + C = \frac{2}{3}x^3 + \frac{1}{2}x^2 - 3x + C. \end{aligned}$$

6.

$$\begin{aligned} \int \frac{(x^4 + 1)^2}{x^3} dx &= \int \frac{x^8 + 2x^4 + 1}{x^3} dx = \int (x^5 + 2x^1 + x^{-3}) dx \\ &= \frac{x^6}{6} + 2 \cdot \frac{x^2}{2} + \frac{x^{-2}}{-2} + C = \frac{1}{6}x^6 + x^2 - \frac{1}{2x^2} + C. \end{aligned}$$